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Part 1: Connection Time Experiment

```
...mship/hydroficient-project -- mosquito -c both_conf.txt -v ... -- -h localhost -p 8883 --cafile certs/ca.pem -t test/topic -d ... ..documents/IoT-Externship/hydroficient-project -- -zah ~/documents/IoT-Externship/hydroficient-project -- -zah +

Last login: Mon Apr 13 22:59:07 on ttys002
kaylacolletti@Mac hydroficient-project % python3 mtl5_benchmark.py --mode connection
mtls Benchmark: Connection Time

Testing TLS...
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/mtls_benchmark.py:23: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client()
TLS Trial 1: 23.1 ms
TLS Trial 2: 20.8 ms
TLS Trial 3: 18.0 ms
TLS Trial 4: 19.1 ms
TLS Trial 5: 18.3 ms
TLS Trial 6: 19.1 ms
TLS Trial 7: 19.0 ms
TLS Trial 8: 17.8 ms
TLS Trial 9: 17.2 ms
TLS Trial 10: 18.7 ms
TLS Trial 11: 19.1 ms
TLS Trial 12: 17.6 ms
TLS Trial 13: 17.9 ms
TLS Trial 14: 18.8 ms
TLS Trial 15: 21.0 ms
TLS Trial 16: 19.2 ms
TLS Trial 17: 19.4 ms
TLS Trial 18: 21.0 ms
TLS Trial 19: 18.9 ms
TLS Trial 20: 19.8 ms

Testing mTLS...
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/mtls_benchmark.py:57: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client()
mTLS Trial 1: 22.7 ms
mTLS Trial 2: 22.7 ms
mTLS Trial 3: 21.8 ms
mTLS Trial 4: 21.3 ms
mTLS Trial 5: 19.4 ms
mTLS Trial 6: 21.0 ms
mTLS Trial 7: 22.1 ms
mTLS Trial 8: 21.4 ms
mTLS Trial 9: 21.4 ms
mTLS Trial 10: 21.4 ms
mTLS Trial 11: 21.1 ms
mTLS Trial 12: 24.7 ms
mTLS Trial 13: 21.9 ms
mTLS Trial 14: 21.8 ms
mTLS Trial 15: 22.1 ms
mTLS Trial 16: 28.3 ms
mTLS Trial 17: 22.8 ms
mTLS Trial 18: 21.1 ms
mTLS Trial 19: 22.1 ms
mTLS Trial 20: 28.9 ms

--- RESULTS ---
TLS avg: 19.2316655088961 ms
mTLS avg: 21.66189571822987 ms
kaylacolletti@Mac hydroficient-project %
```

Metric	One-Way TLS	Mutual TLS
Average latency	19.23 ms	21.66 ms
Min latency	17.2 ms	19.4 ms
Max latency	23.1 ms	24.7 ms
Overhead	--	+2.43 ms (+12.63 %)

Part 2: Message Latency Experiment

Paste your screenshot of the latency benchmark output below:

```
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/mtls_benchmark.py:140: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
pub = mqtt.Client()
Testing 50 messages on port 8883...
Testing 50 messages on port 8884...

--- LATENCY RESULTS ---
=====
Message Latency Comparison
=====
Messages: 50

One-Way TLS:
Average: 0.30 ms
Min: 0.16 ms | Max: 1.46 ms

Mutual TLS:
Average: 0.29 ms
Min: 0.16 ms | Max: 0.96 ms

Overhead: +-0.01 ms (+-4.4%)
kaylacolletti@Mac hydroficient-project %
```

Message Latency Results

Metric	One-Way TLS	Mutual TLS
Average latency	.30 ms	.29 ms
Min latency	.16 ms	.16 ms
Max latency	1.46 ms	.96 ms
Overhead	--	-.01 ms (+-4.4 %)

Part 3: Analysis

1. Connection Overhead

How many extra milliseconds does mTLS add to connection setup?

About +2.43ms on average compared to one-way TLS

The Grand Marina's devices connect once and stay connected for hours. Does this one-time cost matter?

No it does not matter. Since devices only connect once and remain connected for long periods, the small extra cost of ~2.43ms is reasonable compared to hours of uptime. The added security of mTLS outweighs this minor initial delay.

2. Message Latency

How many extra milliseconds does mTLS add per message?

-0.01ms which is very close to 0, meaning mTLS was actually slightly faster than the normal version

Sensor readings are sent every 5 seconds (5,000 ms). What percentage of that interval is the mTLS overhead?

Overhead: 0.01ms

Interval= 5000ms

$(0.01/5000) \times 100 = 0.0002\% \sim 0\%$